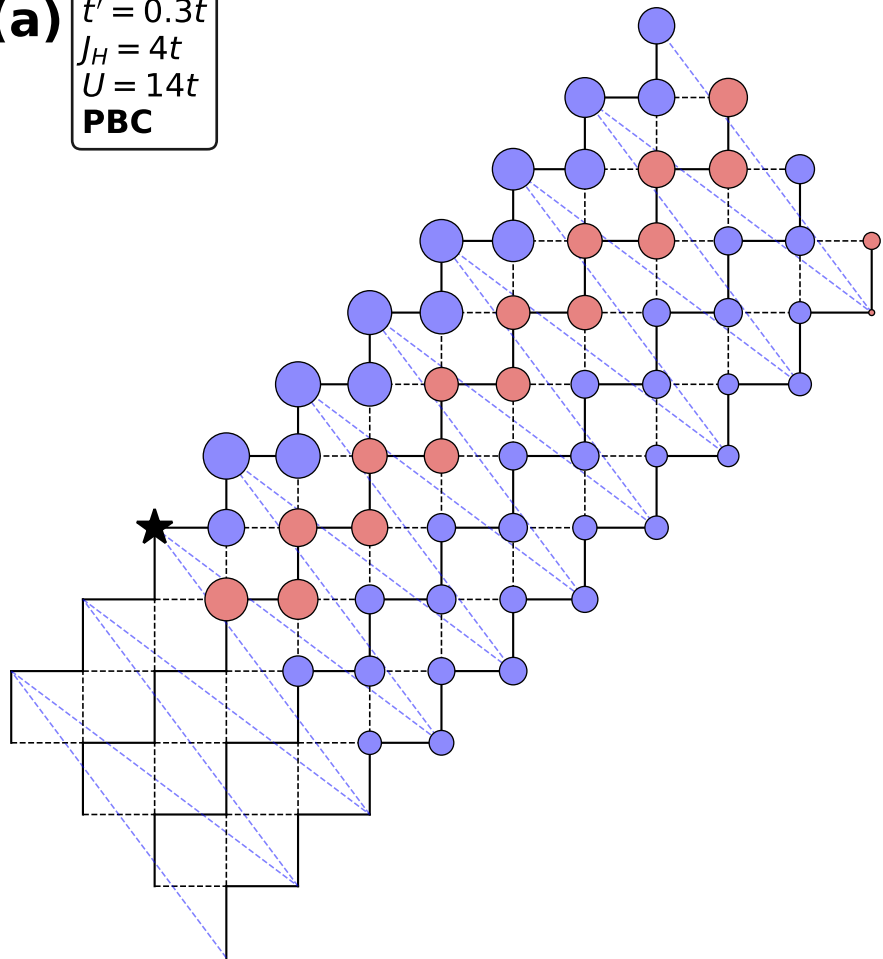
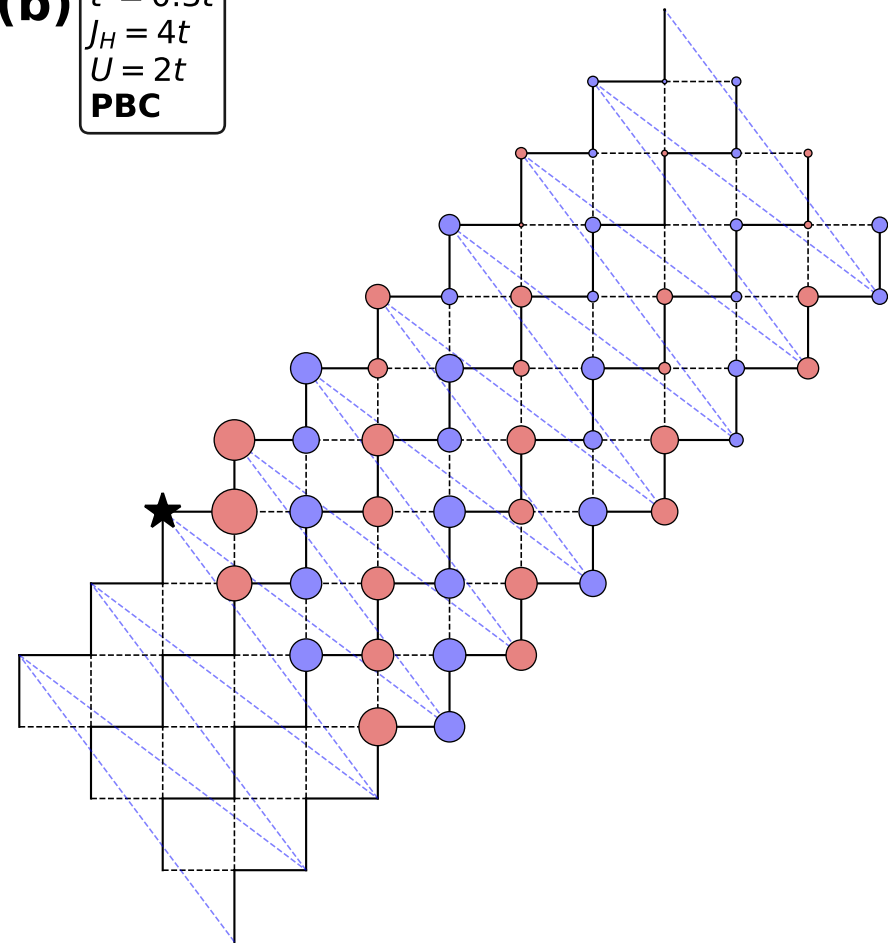


(a)

$$\begin{aligned}t' &= 0.3t \\ J_H &= 4t \\ U &= 14t \\ \text{PBC}\end{aligned}$$

**(b)**

$$\begin{aligned}t' &= 0.3t \\ J_H &= 4t \\ U &= 2t \\ \text{PBC}\end{aligned}$$

 $\langle \mathbf{S}_i \cdot \mathbf{S}_j \rangle$  $+0.059$  -0.059  $+0.0295$  -0.0295  $+0.0059$  -0.0059